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FEATURES

Over the past decade, my work has centred on partnering with teams to build ambitious products for the web across both desktop and mobile. This has provided a ring-side seat to a sweeping variety of teams, products, and technology stacks across more than 100 engagements.

While I'd like to be spending most of this time working through improvements to web APIs, the majority of time spent with partners goes to remediating performance and accessibility issues caused by "modern" frontend frameworks and the culture surrounding them. Today, these issues are most pronounced in React-based stacks.

This is disquieting because React is legacy technology, but it continues to appear in greenfield applications.

Surprisingly, some continue to insist that React is "modern." Perhaps we can square the circle if we understand "modern" to apply to React in the way it applies to art. Neither demonstrate contemporary design and construction. They are not built for current needs and do not meet contemporary performance standards, but pose as expensive objets harkening back to an earlier era's antiquated methods.

The Rule Of Least Client-Side Complexity

OK, But What, Then?

And Nothing Of Value Was Lost

E-Commerce

"But…"

"…we need to move fast"

"…it works for Facebook"

"…our teams already know React"

"…we need to be able to hire easily"

"…everyone has fast phones now"

"… React is industry-standard"

"…the ecosystem…"

"… Next.js can be fast (enough)"

"… React Native!"

In the hope of steering the next team away from the rocks, I've found myself penning advocacy pieces and research into the state of play , as well as giving talks to alert managers and developers of the dangers of today's frontend orthodoxy.

In short, nobody should start a new project in the 2020s based on React. Full stop. 1

The Rule Of Least Client-Side Complexity

Code that runs on the server can be fully costed. Performance and availability of server-side systems are under the control of the provisioning organisation, and latency can be actively managed.

Code that runs on the client, by contrast, is running on The Devil's Computer. 2 Almost nothing about the latency, client resources, or even API availability are under the developer's control.

Client-side web development is perhaps best conceived of as influence-oriented programming . Once code has left the datacentre, all a web developer can do is send thoughts and prayers.

As a direct consequence, an unreasonably effective strategy is to send less code. Declarative forms generate more functional UI per byte sent. In practice, this means favouring HTML and CSS over JavaScript, as they degrade gracefully and feature higher compression ratios. These improvements in resilience and reductions in costs are beneficial in compounding ways over a site's lifetime.

Stacks based on React, Angular, and other legacy-oriented, desktop-focused JavaScript frameworks generally take the opposite bet. These ecosystems pay lip service the controls that are necessary to prevent horrific proliferations of unnecessary client-side cruft. The predictable consequence are NPM-amalgamated bundles full of redundancies like core-js , lodash , underscore , polyfills for browsers that no longer exist, userland ECC libraries, moment.js , and a hundred other horrors.

This culture is so out of hand that it seems 2024's React developers are constitutionally unable to build chatbots without including all of these 2010s holdovers, plus at least one chonky MathML or TeX library in the critical path to display an . A tiny fraction of query responses need to display formulas — and yet.

Tech leads and managers need to break this spell. Ownership has to be created over decisions affecting the client. In practice, this means forbidding React in new work.

OK, But What, Then?

This question comes in two flavours that take some work to tease apart:

The narrow form:

"Assuming we have a well-qualified need for client-side rendering, what specific technologies would you recommend instead of React?"

ECC

Latency (navigation)

The broad form:

A New Era of Creativity: Expert-in-the-loop Generative AI at Stitch Fix

Stitch Fix Multithreaded

Generative AI has been gaining attention and popularity in recent years. Made possible with advances in deep learning algorithms and trained with previously unimaginable amounts of data, generative AI has already contributed many real-world use cases, from creating realistic images with systems like DALL-E 2 and Midjourney, to generating human-like responses with ChatGPT.

At Stitch Fix, we are constantly exploring innovative ways to utilize the latest advancements in AI and ML to enhance the experiences of our clients. In this blog post, we will delve into our approach to generative AI, with a special focus on our text generation use cases. By combining algo-generated text with a human expert-in-the-loop approach, we aim to streamline tasks such as crafting engaging advertisement headlines and producing high-fidelity product descriptions.

Algo-generated Ad Headlines

Generative AI in the text space is powered by large language models (LLMs) that are pre-trained on vast amounts of data (for example, GPT-3 is pre-trained on nearly the entire internet) and can understand and generate natural language. However, once pre-trained, it can generalize from very limited amounts of data and is capable of performing a wide variety of natural language tasks such as Q&A, translation, summarization, and text generation. This few-shot learning capability, which relies only on a few examples to make predictions, makes it especially well suited for tasks that require creativity and originality, such as crafting compelling ad headlines.

Ad headlines are often the first interaction with potential clients, so it's crucial to make them engaging. Traditional marketing requires a copywriter to write new headlines for every new ad asset, which can be time consuming and costly, and may not always result in unique copy. Using generative AI, such as GPT-3, we can quickly generate a large number of headlines tailored to our brand tone and messaging.

We achieve this by using a combination of latent style understanding (check out this blog and this blog to learn more about how we understand clients' personal styles), word embeddings (read more about word embeddings in this blog and this blog), and few-shot learning.

Our ad assets primarily consist of many outfit images that illustrate a wide range of styles we offer. First, we map the outfit and a set of style keywords (such as effortless, classic, romantic, professional, boho etc.) to the latent style space, and then find the style keywords closest to the outfit in that space. Next, we use GPT-3 to generate headlines based on the selected style keywords. Our human experts (copywriters) then review and edit the headlines generated by AI to ensure they capture the style of the outfit and align with the brand's tone and messaging. This human expert-in-the-loop approach allows us to leverage the creativity and efficiency of generative AI while still maintaining human oversight. For our copywriters, reviewing headlines instead of writing new ones saves a significant amount of time and effort.

We have since used the expert-in-the-loop approach for generating all ad headlines for Facebook and Instagram campaigns, and have continued to benefit from improved efficiency while not sacrificing quality. Additionally, we've been constantly seeking opportunities to extend this capability to other areas.

Algo-generated Product Descriptions

"A smart choice for everyday wear, this crew neck T-shirt is a versatile addition to your wardrobe, pairing well with jeans, leggings and shorts."

With the success of algo-generated ad headlines, we have gained confidence in the potential of expert-in-the-loop generative AI for production use cases. Next, we set our sights on a problem of even greater scale and complexity: product descriptions.

Product descriptions play a crucial role in e-commerce and fashion retail websites. Well-written, accurate, and detailed descriptions can enhance the client experience, build trust, and improve search engine optimization. Our Freestyle offering, where clients can shop for individual items in their own personal shopping feed, benefits greatly from informative and compelling product descriptions on the product detail pages (PDP). Writing descriptions for hundreds of thousands of styles in inventory is a daunting task for human copywriters alone, and relying on the few-shot learning approach used for ad headlines results in generic, limited quality descriptions. We needed a solution that could build on the generic large language model with few-shot learning and utilize expert-written examples to create a customized solution for our use case.

Enter, fine-tuning!

Fine-tuning is the process of retraining a pre-trained base model on a smaller, task-specific dataset in order to adapt it to a specific use case. Through fine-tuning, the model learns the unique language, style, and requirements of the specific task, leading to improved performance compared to a generic pre-trained model. For our product description use case, we gathered a task-specific dataset by having our human experts write several hundred high-quality product descriptions (the "completion", or training output) based on product attributes (the "prompt", or training input). We then fine-tuned the base model on this task-specific dataset to teach the model our language, style, and template for high-quality product descriptions. This resulted in accurate and engaging descriptions tailored to meet the needs of our clients, all written in the Stitch Fix brand voice.

Our fine-tuned algo solution offers unbeatable time savings as well as excellent scalability without sacrificing quality of descriptions. In fact, in a blind evaluation, where we com-

Ariadne: building a custom observability UI for personalized search

Stitch Fix Multithreaded

In June 2022, Stitch Fix launched the Freestyle search feature, transforming how our clients discover styles that are tailored to their taste. Under the hood, personalized search is fulfilled by a pipeline of modular microservices. We designed the search system to be composable, making it easy to swap in new components and speeding up experimentation and development. Composability also made the system more observable, making it simpler to make each event traceable and reproducible with built-in logging and metrics. However, even with extensive logging, debugging a complex system such as personalized search can be quite challenging.

As a hypothetical example, if a client searches for running shoes and instead sees

paisley sweaters, we need to quickly diagnose where in the search pipeline the problem occurred,

so that the experts in that system can work on a resolution. Where did the sweaters come from,

and where are the shoes we expected to see instead? Did the query get parsed into incorrect

attributes? Did the attributes get poorly matched with items in our inventory? Did the result

rankings disproportionately focus on the client's past preference for sweaters and disinterest

in sportswear? Understanding which service is responsible for the problem is a crucial first

step in addressing it. Ideally, we want to anticipate and prevent such problems through

interrogating hypothetical search scenarios before clients encounter them.

Early on, we knew that slogging through system logs to understand bugs and dependencies of search would not sustain our need for iterative experimentation and collaborative development. To reduce toil and enhance the impact of the Search team, we invested in building a dedicated tool. Ariadne, named after the labyrinth expert from Greek mythology, is a custom interactive UI designed for search introspection. Ariadne is built with React and custom visualization components based on D3.js, visx, and internal libraries, and is powered by production and experimental search APIs. Developing alongside search, Ariadne has assisted our data scientists, engineers, UX researchers, designers, and product managers through launch, development, and continued experimentation.

In developing Ariadne, we have prioritized several capabilities needed to support and amplify the work of the Search team. In this article, we will show how Ariadne fulfills the following user needs:

Provide a holistic view of the complete search pipeline from query to results;

Make it easy to identify the pipeline component that is "responsible" for anomalous results;

Explore search results visually and by comparing metric distributions;

Focus on specific items and their journey through the search pipeline (e.g. if an expected item is not present in the search results, pinpoint where and why did it get blocked);

Enable "what if" analysis by modifying the search query, client, pipeline configuration and settings, and/or specific inventory items.

Search pipeline from query to results

Search engineers start with searching for a query (e.g. "plaid shacket") for a particular client. The landing page shows how the search pipeline, uniquely configured for that client, transforms the query into search results. Several interactively linked visualizations of the processes and results and high information density support quick triage and analysis

The Sankey (aka flow) diagram at the top of the landing page shows how the search query is handled by each service in the pipeline. When problems arise, this view helps diagnose which service in the pipeline is at the root of the issue. While we use many pipeline configurations at Stitch Fix, the basic pipeline is composed of three services: querious, the query parsing service, elastic, the core search engine backed by ElasticSearch, and tailor-sift, the personalized reranking service.

Sankey diagram section, left. Search query "plaid shacket" is parsed into weighted subqueries recognized by the querious service: "[291] full_text: plaid", "[291] full text: shacket", "[345] gpc_class: Upper Body Wear/Tops", "[345] print_name: Plaid". The numbers in brackets indicate entity weights. Other recognized entities with lower weights are hidden by default.

First, the search query "plaid shacket" is parsed by querious into weighted subqueries (named entities), e.g. "client comments: plaid", "print name: Plaid", "gpc class: Upper Body Wear / Tops" ("shacket" is a fashion term referring to a combination of a shirt and a jacket). The width of each link from the search query to each subquery represents the weight of the subquery. This section of the Sankey diagram helps identify unexpected or improper weighting or errors in named entity recognition. For example, if search results contained paisley sweaters and we saw "print name: Paisley" in this diagram with a high weight, that would indicate that querious improperly identified paisley, a wildly different print from plaid, as highly relevant.

Soaring gas prices and disrupted supply chains will ripple out to increase costs in every store and sector of the economy

Vidya Mani, Associate Professor of Business Administration, University of Virginia; Cornell University · The Conversation US

Americans are already seeing higher gas prices, but that's just the beginning. AP Photo/Carolyn Kaster

The disruptions from the U. S. and Israeli attacks on Iran spread quickly to commercial aircraft, shipping lanes and the world's energy supply . Those repercussions have already hit fuel costs, including for motorists, truckers and fishermen, and are set to spread even more widely , to packaging, household goods, appliances, medicines and electronics.

I study global supply chains and how they interconnect and depend on each other around the world. There are several ways in which U. S. consumers will begin to feel the pinch of the war. Some of those effects have to do with domestic commerce, and some are a result of the interwoven nature of global trade , where raw materials from one place are shipped somewhere they are manufactured into specific items that are then transported to consumers.

Many products are shipped by truck in the U. S., and diesel fuel is more expensive now. Justin Sullivan/Getty Images

Rising costs in the US

There are three main categories in which costs will begin to rise.

Fuel shortages and freight surcharges : From March 2-16, 2026, the average nationwide price of U. S. regular gasoline rose from US$3.01 to $3.96 per gallon , while diesel fuel rose from $3.89 to $5.37. Diesel prices matter to consumer costs because diesel engines power trucks, farm machines, construction equipment, fishing vessels and many of the vehicles that carry domestic freight. When items become more expensive to harvest, build and ship, diesel costs spread quickly into grocery, household and building material prices.

Chemicals, fertilizer and packaging : QatarEnergy has said Iranian attacks on the world's largest liquefied natural gas export plant at Ras Laffan and another plant in Mesaieed, both in Qatar, forced the company to stop producing LNG and associated products on March 2. Two days later, the company declared that it could not fulfill its contracts due to extreme external pressures that would require many years to recover from. The affected products included urea, polymers and methanol, used to make fertilizer, plastics, detergents, packaging and other consumer goods. Reduced production and closed transit routes are also affecting supplies of aluminum and helium produced in the Gulf countries .

Factory slowdowns abroad : When shipping slows and energy costs rise, factories abroad face higher operating costs. As a result they ration production, diverting energy supplies to producing a narrow range of high-value products that can absorb these costs. Diversions of shipment traffic and fewer transportation routes lead to delivery delays. Economic research shows that shipping-cost increases also raise import prices, producer costs and consumer inflation .

Air cargo and delivery delays : Early in the conflict, several countries, including Qatar, Bahrain, Kuwait and the United Arab Emirates, closed their airspace to all traffic . Later advisories warned of risks to planes over neighboring countries as well, except for limited corridors. Those closures affected 20% of global air cargo capacity , raising the risk of delays for higher-value cargo such as medicines, aircraft components and electronics.

Global disruptions

About 80% of the oil and 90% of the LNG moving through the Strait of Hormuz, between the Persian Gulf and the Gulf of Oman, is destined for Asian markets. With strait shipments stopped, consumer electronics and manufacturing hubs in China, Japan, Taiwan and South Korea are drawing on their energy reserves and inventories . But those supplies will run out in a few months. Reduced manufacturing capacity can be expected to cause shortages and higher costs for textiles, chemicals, consumer goods, electronics, appliances, auto parts and fertilizer-intensive industries.

Europe is less directly dependent than Asia on Hormuz shipments, but it is still vulnerable to high LNG prices, increased shipping costs and diesel fuel shortages. Europe has also already faced shortages of heating oil and other fuels as a result of Russia's war on Ukraine . The strait carried about 7% of Europe's LNG inflows in 2025 , and higher costs for energy, ship fuel, freight and insurance can ripple through global trade. For the U. S., that matters because Europe supplies industrial equipment, precision components, medical technology and specialty chemicals sold to businesses and directly to consumers.

African economies are especially exposed to fuel and fertilizer shocks. Large volumes of fertilizer pass through Hormuz , and higher energy and fertilizer prices threaten crop yields and food systems across most of Africa. As a result, U. S. prices can rise for coffee and chocolate – much of which originates in Africa – as well as critical minerals for electric vehicles, energy storage and high-tech equipment.

Grocery prices are affected by costs of fuel and fertilizer. Joe Raedle/Getty Images

Coming home to Americans

This war is not a distant geopolitical shock for U. S. households. It reaches everyday life through fuel, freight, fertilizer, petrochemicals and global supply chains through factories that produce consumer goods.

Some mitigation is possible: 32 nations will be releasing more than 400 million barrels of oil to the global market

Freedom House Ambulance Service

Roman Mars · 99% Invisible

One night halfway through a graveyard shift at the hospital, orderly John Moon watched as two young men burst through the doors. They were working desperately to save a dying patient. Maybe today he wouldn't bat an eye at this scene, but in 1970 nothing about it made sense. The two men weren't doctors, and they weren't nurses. And their strange uniforms weren't hospital issued. Moon was witnessing the birth of a new profession—one that would go on to change the face of emergency medicine. The two men were some of the worlds first paramedics, and, like Moon, they were Black. This is the story of Freedom House Ambulance Service of Pittsburgh. They were the first paramedics and they changed the way we think about emergency medicine.

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

Dear John and Roman

john green · 99% Invisible

Last year, Roman Mars teamed up with Hank Green to guest host Dear Hank & John – this year he's back on the Greens' show once again, but this time with Hank's brother John Green (Turtles All the Way Down, The Fault in Our Stars, The Anthropocene Reviewed).

In their podcast Dear Hank & John , " hosts John and Hank Green (who are also best-selling authors and pioneering YouTubers) offer both humorous and heartfelt advice about life's big and small questions. They bring their personal passions to each episode by sharing the week's news from Mars (the planet) and AFC Wimbledon (the third-tier English football club)."

This week, guest host Roman Mars joins the show to discuss things like: Are roaches a moral failing? How do they do surgery on a fish? Why do only old people like stinky cheese?

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

IN BRIEF

GoCardless Blog, GoCardless Blog: ADDACS messages are the messages you receive from your customer's bank if a mandate has been cancelled or amended by your customer.

JEANNINE BETTS, MARKETING SPECIALIST @ TOPTAL, Toptal Engineering: Trust is an increasingly important factor in consumer purchasing choices. A digital marketing specialist presents a content framework that helps businesses build trust and drive sustainable growth.

Jayson De Leon, Lasha Madan, Jeyca Maldonado Medina, Kelly Prime, 99% Invisible: Welcome to our second episode of short stories all about what may be the original designed object: the trail. If you haven't heard the first episode yet you should totally go back and listen. It's a lot of fun.

Take this episode with you on your next hike!

Trail Mix: Track Two

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

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SiriusXM Podcasts and Roman Mars, 99% Invisible: In 1860, a chance find at sea forever changed our understanding of marine habitats, sparking an unprecedented push to explore a new world of possibilities far below the surface of our planet's oceans. Deep sea life, previously thought possible down ... Continue reading →

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SiriusXM Podcasts and Roman Mars, 99% Invisible: To this day, architects tend to turn their noses up at Las Vegas, or simply dismiss it as irrelevant to serious design theory. But as Denise Scott Brown discovered in the mid-1960s, there is so much to learn from Las Vegas about how to make architecture that speaks to people and not just to architects.

Lessons from Las Vegas

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[redacted]

The Divided Dial

Katie Thornton · 99% Invisible

If you've ever flipped through the radio dial — not satellite, not podcasts, but good old-fashioned AM and FM radio — you may have noticed something. Right wing radio talk is everywhere.

But the airwaves weren't always so dominated by such a narrow range of voices. Reporter and friend of the show Katie Thornton has the story of how talk radio has evolved (and perhaps devolved at times) over the past century, and what all of it means for the airwaves today.

Hear the rest of the the series from On the Media

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

Chapter 4: The List

Roman Mars · 99% Invisible

When Tulicia Lee tried to get help with housing, she was essentially put on a big long list with a bunch of other homeless people. If you live in the U. S., your community probably has a list like this too. Where one ends up on the list can have huge implications, but how one rises to the top of it is a bit of a mystery. In this episode, Katie finally gets to see how it works.

The way homelessness has exploded in California over the last decade, you'd think there was no system in place to address it. But there is one - it just wasn't designed to help everyone. According to Need is a documentary podcast in 5 chapters from 99% Invisible that asks: what are we doing to get people into housing?

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

The Green Book redux

SiriusXM Podcasts and Roman Mars · 99% Invisible

The new film "Green Book" is rolling out across the country. I have not seen the film, so I can't speak to its merits or shortcomings, but while people are possibly being introduced to the concept of the Green Book for the first time, we thought we'd re-release this story from a few years ago about the origin and significance of the Green Book: the Negro Motorists' Travel Guide to the segregated US.

As a special bonus to our story, we also have a Green Book story from Nate DiMeo of the memory palace . Nate had coincidentally written his episode called "Open Road" and we both released them without having heard the other. I think hearing them one after the other is real treat.

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

Stuccoed in Time

Roman Mars, Emmett FitzGerald, Delaney Hall, Katie Mingle · 99% Invisible

Santa Fe is famous in part for a particular architectural style, an adobe (mudbrick) look that came to be called Pueblo Revival. This aesthetic combines elements of indigenous pueblo architecture and the New Mexico's old Spanish missions, resulting in mostly low, brown buildings with smooth edges. Buildings in the city's historical districts in particular have to follow a number of design guidelines so that they fit this desired look; deviating from those aesthetics can stir up a lot of controversy. But this adherence to a single style hasn't always been the norm – for a time, there was actually a powerful push to "Americanize" the city's built environment. Then, over a century ago, a group of preservationists laid out a vision for the look and feel of Santa Fe architecture, and in the process changed the city forever.

Stuccoed in Time

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

Freedom House Ambulance Service: American Sirens

Kevin Hazzard · 99% Invisible

When people ask me what my favorite episode of 99% Invisible is, I have a hard time answering. Not because they're all my precious little babies or some such nonsense, but mostly it's because I just can't remember them all and there's no simple criteria to judge them against each other. But the show is definitely in contention for the best episode we've ever made. It just has everything– engaging storytellers, brilliant reporting, and a compelling history of a moment when the world really changed. It's called the Freedom House Ambulance Service. It originally aired in the summer of 2020, when a lot of the fundamental aspects of work, life, health, law enforcement, structural racism, cities were all being questioned by more and more people because of COVID and the George Floyd protests. Kevin Hazzard, who reported the piece, subsequently released a whole book on the Freedom House Ambulance Service called American Sirens: The Incredible Story of the Black Men Who Became America's First Paramedics . It's new, it's out now, you should buy it. should read it, it should be on all your Christmas lists. To celebrate the book's release, I'm proud to re-present to you: The remarkable story of the Freedom House Ambulance Service.

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

The Power Broker #11: Brennan Lee Mulligan

elliott kalan, isabel angell · 99% Invisible

This is the eleventh official episode, breaking down the 1974 Pulitzer Prize winning book, The Power Broker by our hero Robert Caro.

This week, Roman Mars and Elliott Kalan sit down with Brennan Lee Mulligan, a comedian and host with Dropout TV, where he's the creator of Dimension 20 — a Dungeons & Dragons show that features incredibly complex and campaigns, with improv actors and special effects. And as the Dungeon Master, Brennan leads these stories. Season three of Dimension 20 takes place in a magical New York City, where the main villain is a fictionalized, undead Robert Moses, who shares the real Robert Moses's passion for building roads and destroying lives through bureaucracy.

Elliott and Roman also cover the second section of Part 7 (Chapter 42 through Chapter 46), discussing the major story beats and themes.

The Power Broker #11: Brennan Lee Mulligan

Join the discussion on Discord and Reddit .

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

It's called the Freedom House Ambulance Service.

Kevin Hazzard

Empire of the Sum

keith houston, lasha madan · 99% Invisible

Keeping track of numbers has always been part of what makes us human. So at some point along the way, we created a tool to help us keep count, and then we gave that tool a name. We called it: a calculator. But depending on what era you were born in, and maybe even what country, what constituted a 'calculator' varied widely.

Keith Houston wrote about the evolution of the calculator in his latest book, Empire of the Sum The Rise and Reign of the Pocket Calculator . It is exactly the kind of nerdery we like to get up to here at 99% Invisible – history explained through the lens of an everyday designed object.

Empire of the Sum

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

Airships and the Future that Never Was

SiriusXM Podcasts and Roman Mars · 99% Invisible

They are hulking, but graceful – human-made whales that float in the air. For over a century, lighter-than-air vehicles have captured the public imagination, playing a recurring role in our dreams of alternate realities and futures that might have been. In these visions, cargo and passengers traverse the globe in smoothly gliding aircraft, then dock elegantly at the mooring towers on top of Art Deco skyscrapers.

Today, blimps are mostly just PR gimmicks, but for 100 years, lighter-than-air crafts were seriously considered as the perfect design solution for all kinds of problems, at least in theory. And despite setbacks and failures, people just wouldn't give up on the promise of airships.

The most promising (and most opulent) rigid airship of the 1920s era was Britain's R101 (the R stands for rigid) and its rise and dramatic fall is the primary subject of engineering expert Bill Hammack's new book about Britain's last great airship, called Fatal Flight.

Airships and the Future that Never Was

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

The Lost Subways of North America

Jake Berman, Sarah Baik, Kelly Prime · 99% Invisible

Los Angeles actually used to have a massive electric railway system in the early 1900s, called the Red Car. Jake Berman, the author of The Lost Subways of North America , tells us about how, time after time, when North American cities seemed just inches away from having a robust, utopian future of fast, reliable, and convenient public transportation systems, something gets in the way. That thing is sometimes dysfunctional local politics, sometimes it's bureaucracy. Sometimes it's the way our infrastructure favors cars over mass transit, and too often, it's racism.

The Lost Subways of North America

Start a free trial now on Apple Podcasts or by visiting siriusxm.com/podcastsplus .

Exploring The 99% Invisible City

Roman Mars · 99% Invisible

We're excited to celebrate the release of The 99% Invisible City book by host Roman Mars and producer Kurt Kohlstedt with a guided audio tour of beautiful downtown Oakland, California.

In this episode, we explain how anchor plates help hold up brick walls; why metal fire escapes are mostly found on older buildings; what impact camouflaging defensive designs has on public spaces; who benefits from those spray-painted markings on city streets, and much more.

Plus, At the end of the tour, stick around for a behind the scenes look at the book as we answer a series of fan-submitted questions about how it was created, offering a window into the writing, illustration and design processes.

Exploring The 99% Invisible City

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